

Show that the following inequality holds for all $a, b, c > 0$.

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$$(a^3 + 2)(b^3 + 2)(c^3 + 2) \geq (a + b + c)^3$$

$$a_i, \dots, z_i \in \mathbb{R}^+, \lambda_a, \dots, \lambda_z \geq 0 \text{ s.t. } \lambda_a + \dots + \lambda_z = 1$$

$$\left(\sum_{i=1}^n a_i \right)^{\lambda_a} \dots \left(\sum_{i=1}^n z_i \right)^{\lambda_z} \geq \sum_{i=1}^n a_i^{\lambda_a} \dots z_i^{\lambda_z}$$

$$\left((a^3 + 1 + 1)^{\frac{1}{3}} (1 + b^3 + 1)^{\frac{1}{3}} (1 + 1 + c^3)^{\frac{1}{3}} \right)^3 \geq (a + b + c)^3$$

$$(a^3 + 1 + 1)(1 + b^3 + 1)(1 + 1 + c^3) \geq (a + b + c)^3$$

□